

EXPERIMENTAL DATA ON MINIMAL NOR GATES CIRCUITS

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1. INTRODUCTION

This paper presents numerical data and statistics related to circuits with a minimum number of NOR gates assuming unlimited fan-off. While faster circuits exist in terms of signal propagation time, they require a larger number of gates. The first results concern circuits computing functions of the form

$$f : \{0, 1\}^n \rightarrow \{0, 1\}$$

while the second results concern circuits computing permutations. We may restrict ourselves to NOR gates since the scope of this paper is about asymptotic behaviors: any 2-input gate can be realized with at most five NOR gates. The data and proof of optimality were obtained by constructing every possible circuit with a fixed number of input wires. The computations were done on an Intel Core i5-6600K CPU along with a NVIDIA GeForce RTX 3080 Ti GPU over the course of a single month. The code is available at <https://github.com/AntoineComeau/norgates>.

2. SINGLE OUTPUT FUNCTIONS

Given a function f as described above, let $N(f)$ the minimum number of NOR gates required to construct a circuit that computes f . Let $M_n(m)$ the number of functions f with n input wires such that $N(f) = m$. Finally, let $B(n)$ be the maximum of $N(f)$ over every function f with n inputs. In particular,

$$\sum_{i=0}^{B(n)} M_n(i) = 2^n.$$

Here are the results for $n = 1, 2, 3, 4$.

$B(n)$	3	5	10	14
n	1	2	3	4

$M_1(i)$	1	1	1	1
i	0	1	2	3

$M_2(i)$	2	3	4	4	2	1
i	0	1	2	3	4	5

$M_3(i)$	3	6	13	26	43	48	53	36	22	5	1
i	0	1	2	3	4	5	6	7	8	9	10

$M_4(i)$	4	10	31	98	293	807	2067	4351	8941	13085	16804	12210	5464	1319	52
i	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14

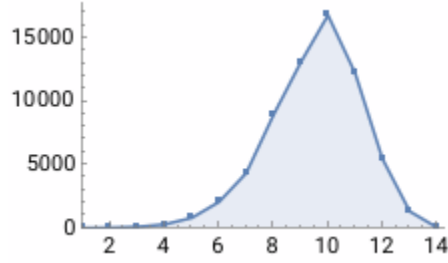


FIGURE 1. Plot of M_4

The function corresponding to $M_1(3)$ is the constant function 1. The function corresponding to $M_2(5)$ is the XOR function. The function corresponding to $M_3(10)$ is the following.

output	1	1	1	0	1	0	0	1
input	000	001	010	011	100	101	110	111

For $M_4(14)$, the following is an example among the 52 candidates.

output	0	1	1	0	1	0	0	1	1	0	0	1	0	1	1	1
input	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111

One possible circuit that computes this function is given by

$$\begin{aligned}
x_4 &= x_0 \text{ NOR } x_1; x_5 = x_3 \text{ NOR } x_2; x_6 = x_4 \text{ NOR } x_0; x_7 = x_4 \text{ NOR } x_1; x_8 = x_5 \text{ NOR } x_2 \\
x_9 &= x_7 \text{ NOR } x_6; x_{10} = x_9 \text{ NOR } x_2; x_{11} = x_{10} \text{ NOR } x_3; x_{12} = x_{11} \text{ NOR } x_4; x_{13} = x_{11} \text{ NOR } x_8 \\
x_{14} &= x_{13} \text{ NOR } x_5; x_{15} = x_{13} \text{ NOR } x_9; x_{16} = x_{14} \text{ NOR } x_{12}; x_{17} = x_{15} \text{ NOR } x_{16}
\end{aligned}$$

where x_0, x_1, x_2, x_3 are the four input wires and x_{17} is the output wire. A faster circuit in terms of signal propagation may exist using 14 NOR gates, but this is outside the scope of this paper. This is the first solution found by the code.

3. PERMUTATIONS

Let $\pi \in S_r$ be a permutation of r elements, where $r = 2^n$. As above, let $N(\pi)$ be the minimal number of NOR gates required to construct a circuit that computes $\pi : \{0, 1\}^n \rightarrow \{0, 1\}^n$. Let also $M_n(m)$ and $B(n)$ be defined as above, but for permutations. In particular,

$$\sum_{i=0}^{B(n)} M_n(i) = 2^n!.$$

Here are the results for $n = 1, 2, 3$.

$B(n)$	1	5	15
n	1	2	3

$M_1(i)$	1	1
i	0	1

$M_2(i)$	2	4	2	0	3	13
i	0	1	2	3	4	5

i	$M_3(i)$
0	6
1	18
2	18
3	6
4	36
5	162
6	252
7	396
8	972
9	3648
10	6792
11	10770
12	10614
13	5922
14	672
15	36

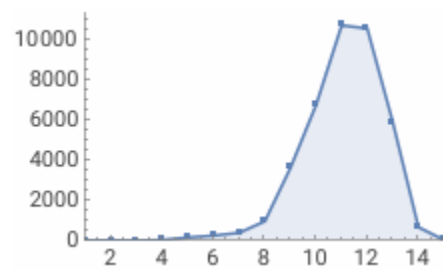


FIGURE 2. Plot of M_3

In the following table, the value at (x, y) is equal to

$$\#\{\pi \in S_4 \text{ such that } N(\pi) = x \text{ and } N(\pi^{-1}) = y\}.$$

$y \backslash x$	0	1	2	3	4	5
0	2	0	0	0	0	0
1	0	4	0	0	0	0
2	0	0	2	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	2	1
5	0	0	0	0	1	12

The single permutation at $(5, 4)$ is given by $(0 \ 2 \ 1)$ under cycle notation.

In the following table, the value at (x, y) is equal to

$$\#\{\pi \in S_8 \text{ such that } N(\pi) = x \text{ and } N(\pi^{-1}) = y\}.$$

$y \backslash x$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	18	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	0	0	18	0	0	0	0	0	0	0	0	0	0	0	0	0
3	0	0	0	6	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	36	0	0	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0	162	0	0	0	0	0	0	0	0	0	0
6	0	0	0	0	0	0	252	0	0	0	0	0	0	0	0	0
7	0	0	0	0	0	0	0	396	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	756	180	36	0	0	0	0	0
9	0	0	0	0	0	0	0	0	180	2496	828	144	0	0	0	0
10	0	0	0	0	0	0	0	0	36	828	3678	1854	360	36	0	0
11	0	0	0	0	0	0	0	0	0	144	1854	5766	2736	270	0	0
12	0	0	0	0	0	0	0	0	0	0	360	2736	5430	2052	36	0
13	0	0	0	0	0	0	0	0	0	0	36	270	2052	3384	180	0
14	0	0	0	0	0	0	0	0	0	0	0	0	36	180	456	0
15	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	36

We observe a strong correlation between a permutation and its inverse, especially when $N(\pi)$ is low. The group $S_3 \times S_3$ acts on S_8 by permuting the input and output wires. This induces a partition of S_8 . The 36 permutations with the highest difference $N(\pi) - N(\pi^{-1}) = 3$ are a single block of this partition, with $\pi = (0 \ 4)(1 \ 2 \ 3 \ 5)(6 \ 7)$ as an example under cycle notation.

The 36 permutations at position $(15, 15)$ are also a single block of this partition. Here's a circuit computing a permutation with $N(\pi) = 15$.

$x_3 = x_0 \text{ NOR } x_0$; $x_4 = x_1 \text{ NOR } x_0$; $x_5 = x_2 \text{ NOR } x_0$; $x_6 = x_4 \text{ NOR } x_2$; $x_7 = x_5 \text{ NOR } x_1$
 $x_8 = x_6 \text{ NOR } x_3$; $x_9 = x_8 \text{ NOR } x_7$; $x_{10} = x_9 \text{ NOR } x_9$; $x_{11} = x_{10} \text{ NOR } x_5$; $x_{12} = x_{10} \text{ NOR } x_6$
 $x_{13} = x_{11} \text{ NOR } x_4$; $x_{14} = x_{12} \text{ NOR } x_8$; $x_{15} = x_{14} \text{ NOR } x_1$; $x_{16} = x_{15} \text{ NOR } x_9$; $x_{17} = x_{16} \text{ NOR } x_4$

where x_0, x_1 , and x_2 are the input wires and x_{13}, x_{14} , and x_{17} are the output wires. Under cycle notation, this circuit computes $(1\ 3\ 6\ 4\ 2\ 7)$.

The main question is about whether the asymptotic behavior of

$$\max_{\pi \in S_n} (N(\pi) - N(\pi^{-1}))$$

is polynomial or exponential.